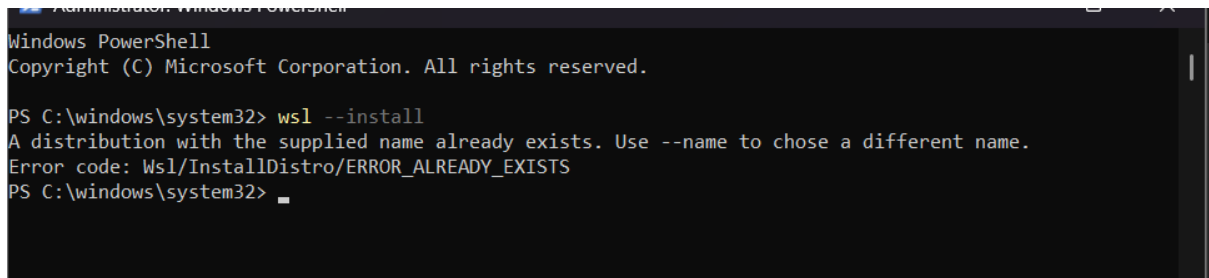


OS_SKILL

- 1) Step 1: Search for Windows Powershell & click on “Run As Administrator”
- 2) Step 2:



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> wsl --install
A distribution with the supplied name already exists. Use --name to chose a different name.
Error code: Wsl/InstallDistro/ERROR_ALREADY_EXISTS
PS C:\windows\system32> _
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for Ubuntu in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

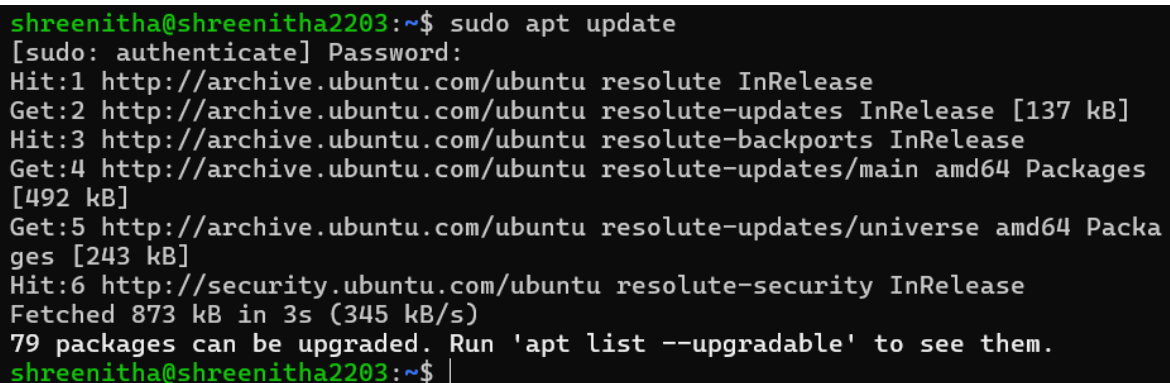
Enter new UNIX username: Shreenitha

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****



```
shreenitha@shreenitha2203:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages
[492 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packa
ges [243 kB]
Hit:6 http://security.ubuntu.com/ubuntu resolute-security InRelease
Fetched 873 kB in 3s (345 kB/s)
79 packages can be upgraded. Run 'apt list --upgradable' to see them.
shreenitha@shreenitha2203:~$ |
```

Step 7: Install GCC (C Compiler)

```
shreenitha@shreenitha2203:~$ sudo apt install build-essential
Installing:
  build-essential

Installing dependencies:
  bzip2                libalgorithm-diff-xs-perl
  dpkg-dev              libalgorithm-merge-perl
  fakeroot              libcrypt-dev
  g++                   libdpkg-perl
  g++-15                libfakeroot
  g++-15-x86-64-linux-gnu libfile-fcntllock-perl
  g++-x86-64-linux-gnu  libstdc++-15-dev
  libalgorithm-diff-perl lto-disabled-list

Suggested packages:
  bzip2-doc                g++-multilib      bzip
  debian-keyring           g++-15-multilib  libstdc++-15-doc
  debian-tag2upload-keyring gcc-15-doc

Summary:
  Upgrading: 0, Installing: 17, Removing: 0, Not Upgrading: 79
  Download size: 19.1 MB
  Space needed: 72.9 MB / 1024 GB available

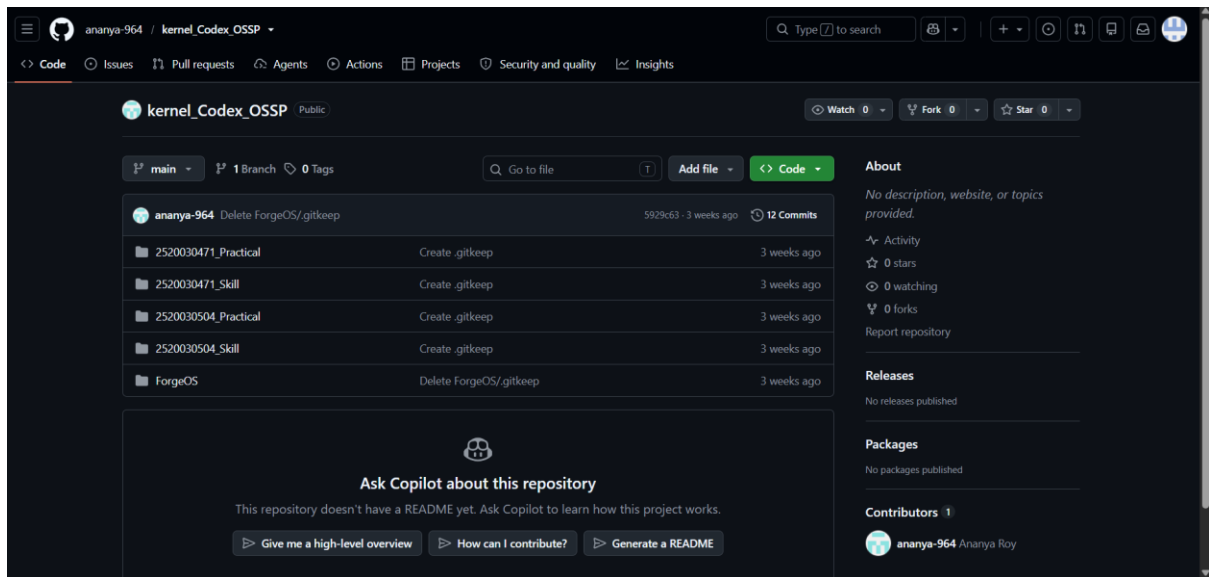
Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libstdc++-15-dev
amd64 15.2.0-16ubuntu1 [2920 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-15-x86-64-lin
ux-gnu amd64 15.2.0-16ubuntu1 [14.4 MB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-15 amd64 15.2
.0-16ubuntu1 [26.7 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-x86-64-linux-
gnu amd64 4:15.2.0-5ubuntu1 [968 B]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++ amd64 4:15.2.
0-5ubuntu1 [1100 B]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdpkg-perl all
1.23.7ubuntu1 [289 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 bzip2 amd64 1.0.8
```

Step 8: Verify the Installation

```
shreenitha@shreenitha2203:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

shreenitha@shreenitha2203:~$ |
```

GITHUB REPOSITORY



1)

```
shreenitha@shreenitha2203:~$ mkdir ossp
shreenitha@shreenitha2203:~$ cd ossp
shreenitha@shreenitha2203:~/oss$ mkdir src
shreenitha@shreenitha2203:~/oss$ ls
src
shreenitha@shreenitha2203:~/oss$ cd src
shreenitha@shreenitha2203:~/oss/src$ pwd
/home/shreenitha/oss/src
```

Fork():

```

#include <stdio.h>
#include <unistd.h>

int main()
{
    int pid;
    pid = fork();
    if (pid == 0)
    {
        printf("Child Process\n");
        printf("Child pid is %d\n", getpid());
    }
    else if(pid>0)
    {
        printf("Parent process\n");
        printf("Parent process pid is %d\n", getpid());
    }
    else if(pid<0)
    {
        printf("Failed process\n");
    }
}

```

```

shreenitha@shreenitha2203:~/ossp/src$ nano fork1.c
shreenitha@shreenitha2203:~/ossp/src$ gcc fork1.c
shreenitha@shreenitha2203:~/ossp/src$ ./a.out
Parent process
Parent process pid is 658
Child Process
Child pid is 659
shreenitha@shreenitha2203:~/ossp/src$ |

```

Exec()

```

#include<unistd.h>
int main()
{
    char *args[]={ "ls", "-l", NULL };
    execvp("ls", args);
}

```

```

shreenitha@shreenitha2203:~/ossp/src$ nano exec2.c
shreenitha@shreenitha2203:~/ossp/src$ gcc exec.c
cc1: fatal error: exec.c: No such file or directory
compilation terminated.
shreenitha@shreenitha2203:~/ossp/src$ gcc exec2.c
shreenitha@shreenitha2203:~/ossp/src$ ./a.out
total 48
-rwxr-xr-x 1 shreenitha shreenitha 16008 Aug 23 14:42 a.out
-rw-r--r-- 1 shreenitha shreenitha 163 Aug 23 10:10 exec1.c
-rw-r--r-- 1 shreenitha shreenitha 84 Aug 23 14:41 exec2.c
-rwxr-xr-x 1 shreenitha shreenitha 16080 Aug 23 10:40 fork1
-rw-r--r-- 1 shreenitha shreenitha 230 Aug 23 10:28 fork1.c
-rw-r--r-- 1 shreenitha shreenitha 33 Aug 23 11:29 shell.c
shreenitha@shreenitha2203:~/ossp/src$ |

```

Ps tree

```

shreenitha@shreenitha2203:~/ossp/src$ gcc exec1.c
shreenitha@shreenitha2203:~/ossp/src$ ./a.out
Before exec
a.out exec1.c fork1.c
shreenitha@shreenitha2203:~/ossp/src$ nano exec1.c
shreenitha@shreenitha2203:~/ossp/src$ nano fork1.c
shreenitha@shreenitha2203:~/ossp/src$ gcc fork1.c
shreenitha@shreenitha2203:~/ossp/src$ ./a.out
Parent: PID=1190
Child: PID=1191 PPID=1190
shreenitha@shreenitha2203:~/ossp/src$ nano fork1.c
shreenitha@shreenitha2203:~/ossp/src$
shreenitha@shreenitha2203:~/ossp/src$ cd ..
shreenitha@shreenitha2203:~/ossp$ pstree -p
systemd(1)─agetty(224)
            └─chronyd-starter(162)─chronyd(259)─chronyd(273)
            └─cron(163)
            └─dbus-daemon(164)
            └─init-systemd(Ub(2))─SessionLeader(314)─Relay(319)(316)─bash(319)─pstree(1282)
                                   └─init(6)─{init}(7)
                                   └─login(322)─bash(414)
                                   └─init-systemd(Ub)(6)
            └─networkd-dispatcher(173)
            └─rsyslogd(220)─{rsyslogd}(260)
                           └─{rsyslogd}(261)
                           └─{rsyslogd}(262)
            └─systemd(382)─(sd-pam)(384)
            └─systemd-journal(46)
            └─systemd-logind(182)
            └─systemd-resolve(78)
            └─systemd-udevd(85)─(udev-worker)(1275)
                                └─(udev-worker)(1276)
            └─unattended-upgr(270)─{unattended-upgr}(294)
            └─wsl-pro-service(1233)─{wsl-pro-service}(1234)
                                    └─{wsl-pro-service}(1235)
                                    └─{wsl-pro-service}(1236)
                                    └─{wsl-pro-service}(1237)
                                    └─{wsl-pro-service}(1238)
                                    └─{wsl-pro-service}(1239)
                                    └─{wsl-pro-service}(1242)
shreenitha@shreenitha2203:~/ossp$ |

```

QUESTION2

```

#include <stdio.h>
#include <string.h>

int main()
{
    char command[50];

    while(1)
    {
        printf("myshell> ");
        scanf("%s", command);

        if(strcmp(command, "exit") == 0)
        {
            printf("Goodbye!\n");
            break;
        }

        printf("You entered: %s\n", command);
    }

    return 0;
}

```

```

shreenitha@shreenitha2203:~/ossp$ nano src/shell.c
shreenitha@shreenitha2203:~/ossp$ gcc src/myshell.c -o myshell
cc1: fatal error: src/myshell.c: No such file or directory
compilation terminated.
shreenitha@shreenitha2203:~/ossp$ gcc src/shell.c
shreenitha@shreenitha2203:~/ossp$ ./a.out
myshell> hello
You entered: hello
myshell> how are you
You entered: how
myshell> You entered: are
myshell> You entered: you
myshell> exist
You entered: exist
myshell> exist
You entered: exist
myshell> exit
Goodbye!

```

```

#include <stdio.h>
#include <string.h>

int main()
{
    char command[50];

    while(1)
    {
        printf("myshell> ");
        fgets(command, 50, stdin);

        command[strcspn(command, "\n")] = '\0';

        if(strcmp(command, "exit") == 0)
        {
            printf("Goodbye!\n");
            break;
        }

        printf("You entered: %s\n", command);
    }

    return 0;
}

```

```

shreenitha@shreenitha2203:~/oosp$ nano src/shell.c
shreenitha@shreenitha2203:~/oosp$ gcc src/shell.c
shreenitha@shreenitha2203:~/oosp$ ./a.out
myshell> how are you
You entered: how are you
myshell> exit
Goodbye!

```